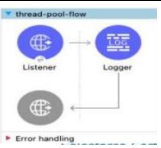
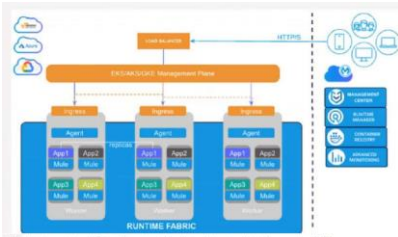
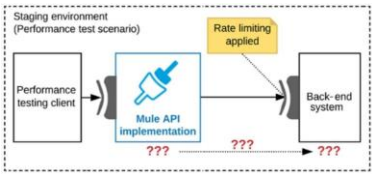


No	Question	Ans
1	An organization has decided on a cloud migration strategy to minimize the organization's own IT resources. Currently, the organization has all of its Mule applications running on its own premises and uses an on-premises load balancer that exposes all APIs under the base URL <code>https://api.acme.com</code>. As part of the migration strategy, the organization plans to migrate all of its Mule applications and the on-premises load balancer to CloudHub 1.0. What is the most straightforward and cost-effective approach to Mule application deployment and load balancing that preserves the public URLs? A. Deploy the Mule applications to CloudHub 1.0. Update the CNAME record for <code>api.acme.com</code> in the organization's DNS server to point to the A record of the CloudHub Shared Load Balancer (SLB). Apply mapping rules in the SLB to map URLs to their corresponding Mule applications. B. Deploy the Mule applications to CloudHub 1.0. Create a CNAME record for <code>api.acme.com</code> in the CloudHub 1.0 Shared Load Balancer (SLB) that points to the A record of the on-premises load balancer. Apply mapping rules in the SLB to map URLs to their corresponding Mule applications. C. For each migrated Mule application, deploy an API proxy Mule application to CloudHub 1.0, with all traffic to the Mule applications routed through a CloudHub 1.0 Dedicated Load Balancer (DLB). Update the CNAME record for <code>api.acme.com</code> in the organization's DNS server to point to the A record of the CloudHub 1.0 DLB. Apply mapping rules in the DLB to map each API proxy application to its corresponding Mule application. D. Deploy the Mule applications to CloudHub 1.0. Update the CNAME record for <code>api.acme.com</code> in the organization's DNS server to point to the A record of a CloudHub 1.0 Dedicated Load Balancer (DLB). Apply mapping rules in the DLB to map URLs to their corresponding Mule applications.	https://docs.mulesoft.com/cloudhub/cloudhub-dedicated-load-balancer D
2	A bank is implementing a REST API in a Mule application to receive an array of accounts from an online banking platform user interface (UI), retrieve account balances for those accounts from a backend Finance system, and then return the account balances so they can be displayed in the online banking platform UI. As part of the processing, the MuleSoft API also needs to insert the retrieved account data into an Audit Database for auditing purposes. The auditing process should not add latency to the account balance retrieval response back to the online banking platform UI. The <code>retrieveBalances</code> flow in the Mule application is designed to use an operation in a connector to the Finance system (the Finance operation) that can only look up one account record at a time, and a operation from a different connector to the Audit system (the Audit operation) that can only insert one account record at a time. To best meet the performance-related requirements, what scope or scopes should be used and how should they be used to incorporate the Finance operation and Audit operation into the <code>retrieveBalances</code> flow? A. Wrap the Finance operation in a Parallel For-Each scope. Wrap the Audit operation in a Async scope. B. Wrap the Finance operation in a Until-Successful scope. Wrap the Audit operation in a Try-Catch scope. C. Wrap both connector operations in a Async scope. D. Wrap both connector operations in a For-Each scope.	D
3	A large life sciences customer plans to use the Mule Tracing module with the Mapped Diagnostic Context (MDC) logging operations to enrich logging in its Mule application and to improve tracking by providing more context in the Mule application logs. The customer also wants to improve throughput and lower the message processing latency in its Mule application flows. After installing the Mule Tracing module in the Mule application, how should logging be performed in flows in Mule applications, and what should be changed in the <code>log4j2.xml</code> files? A. In the flows, add Mule Tracing module Set logging variable operations before any Core Logger components. In <code>log4j2.xml</code> files, change the appender's pattern layout to use <code>%MDC</code> and then assign the appender to a Logger or Root element. B. In the flows, add Mule Tracing module Set logging variable operations before any Core Logger components. In <code>log4j2.xml</code> files, change the appender's pattern layout to use the <code>%MDC</code> placeholder and then assign the appender to an AsyncLogger element. C. In the flows, add Mule Tracing module Set logging variable operations before any Core Logger components. In <code>log4j2.xml</code> files, change the appender's pattern layout to use <code>%asyncLogger</code> placeholder and then assign the appender to an AsyncLogger element. D. In the flows, wrap Logger components in Async scopes. In <code>log4j2.xml</code> files, change the appender's pattern layout to use the <code>%asyncLogger</code> placeholder and then assign the appender to a Logger or Root element.	https://docs.mulesoft.com/mule-runtime/latest/logging-mdc A
4	An organization is building out a test suite for their application using MUnit. The Integration Architect has recommended using Test Recorder in Anypoint Studio to record the processing flows and then configure unit tests based on the captured events. What is a core consideration that must be kept in mind while using Test Recorder? A. The Recorder supports loops where the structure of the data being tested changes inside the iteration B. Mocking values resulting from parallel processes are possible and will not affect the execution of the processors that follow in the test C. The Recorder supports mocking a message before or inside a Foreach processor D. Tests for flows cannot be created if Mule errors are raised inside the flows, even if the errors are handled by On-Error Continue error handlers	https://docs.mulesoft.com/munit/latest/test-recorder D
5	A manufacturing company wants to share inventory updates with dealers D1 and D2 asynchronously and concurrently via Queues Q1 and Q2. Dealer D1 must consume the messages from the Queue Q1, and Dealer D2 must consume the messages from the Queue Q2. Dealer D1 has implemented a retry mechanism to reprocess the transaction in case of any errors while processing the inventory updates. Dealer D2 has not implemented any retry mechanism. How should the dealers acknowledge the message to avoid message loss and minimize impact on the current implementation? A. Dealer D1 can use AUTO acknowledgement and Dealer D2 must use MANUAL acknowledgment and acknowledge the message after successful processing B. Dealer D1 can use AUTO acknowledgement and Dealer D2 can use IMMEDIATE acknowledgment and acknowledge the message after successful processing C. Dealer D2 must use AUTO acknowledgement and Dealer D1 can use MANUAL acknowledgment and acknowledge the message after successful processing D. Dealer D1 and Dealer D2 must use AUTO acknowledgement and acknowledge the message after successful processing	https://docs.mulesoft.com/jms-connector/latest/jms-ack A

6	An architect is designing a Mule application to meet the following two requirements: 1. The application must process files asynchronously and reliably from an FTPS server to a back-end database using VM intermediary queues for load-balancing Mule events. 2. The application must process a medium rate of records from a source to a target system using a Batch Job scope. To make the Mule application more reliable, the Mule application will be deployed to two CloudHub 1.0 workers. Following MuleSoft-recommended best practices, how should the Mule application deployment typically be configured in Runtime Manager to best support the performance and reliability goals of both the Batch Job scope and the file processing VM queues? A. Check the Persistent VM queues checkbox in the application deployment configuration B. Check the Non-persistent VM queues checkbox in the application deployment configuration C. In the Runtime Manager Properties tab, disable persistent VM queues for Batch Job scopes D. In the Runtime Manager Properties tab, enable persistent VM queues for the FTPS connector	A	https://help.salesforce.com/s/articleView?id=001120515&type=1
7	A company is modernizing its legacy systems to accelerate access to applications and data while supporting the adoption of new technologies. The key to achieving this business goal is unlocking the company's systems and data, including a set of existing services hosted on-premises that can be accessed by authorized external clients. The IT staff is mainly experienced only with supporting its legacy systems. Considering the current aggressive backlog and the project delivery requirements, the company wants to take a strategic approach in the first phase of its transformation projects by quickly deploying APIs in Mule runtimes that are able to scale, connect to on-premises systems, and migrate as needed. Following MuleSoft best practices, what MuleSoft runtime deployment option best meets the company's goals to begin its digital transformation journey? A. Runtime Fabric on VMs/bare metal B. CloudHub runtimes C. Customer-hosted runtimes provisioned by a MuleSoft services partner D. Customer-hosted self-provisioned runtimes	A	
8	A gaming company has implemented an API as a Mule application and deployed the API implementation to a CloudHub 2.0 private space. The API implementation must connect to a mainframe application running in the customer's on-premises corporate data center and also to a Kafka cluster running in an Amazon AWS VPC. What is the most efficient way to enable the API to securely connect from its private space to the mainframe application and Kafka cluster? A. In Runtime Manager, set up VPC peering between the CloudHub 2.0 private network and the on-premises data center. In the AWS account, set up VPC peering between the the AWS VPC and the CloudHub 2.0 private network. B. In the AWS account, attach the CloudHub 2.0 private space to an AWS transit gateway that routes from the CloudHub 2.0 private space to the on-premises data center. In Runtime Manager, configure an Anypoint VPN to route from the CloudHub 2.0 private space to the AWS VPC. C. In Runtime Manager, configure an Anypoint VPN to route from the CloudHub 2.0 private space to the on-premises data center. In the AWS account, attach the private space to an AWS transit gateway that routes to the AWS VPC. D. In the AWS account, attach the private space directly to the AWS VPC. In the AWS account, use an AWS transit gateway to route from the AWS VPC to the on-premises data center.	C	https://docs.mulesoft.com/cloudhub-2/ps-vpn-about
9	An insurance company is implementing a new Mule application that supports a set of critical functions driven by a REST-API-enabled claim payment rules engine hosted on Oracle ERP. The new Mule application requires many data transformation operations as it performs batch processing logic. The company wants to leverage and reuse as many of its existing Java-based capabilities (such as classes, objects, frameworks, and data models) as possible. What approach should be used when implementing the required data mappings and transformations between other Mule application components and Oracle ERP? A. Transform by calling any suitable Java class from DataWeave B. Create new metadata RAML classes in the Mule runtime corresponding to existing Java objects, and then perform required transformations via DataWeave C. Invoke any of the appropriate Java methods directly, create corresponding metadata RAML classes, and then perform required transformations via DataWeave D. Transform via the XSLT module	A	https://docs.mulesoft.com/dataweave/latest/dataweave-cookbook-java-methods
10	A healthcare provider is implementing Anypoint Platform to manage its application infrastructure and is using the customer-hosted runtime for its business due to certain financial requirements it must meet. It has built a number of synchronous APIs and is currently hosting these on a Mule runtime on one server. These applications make use of a number of components, including heavy use of Object Stores and VM Queues. Business has grown rapidly in the last year, and the healthcare provider is starting to receive reports of reliability issues from its applications. The DevOps team indicates that the APIs are currently handling too many requests, and this is overloading the server. The team has also mentioned that there is significant downtime when the server is down for maintenance. What can the team add or upgrade to mitigate these issues? A. Add a load balancer and add additional servers in a cluster configuration B. Upgrade the networking hardware of the server C. Add additional servers in a server group configuration D. Upgrade the CPU and memory of the server	A	
11	Refer to the exhibit.  In the flow of this Mule application, what thread pool handles requests to the HTTP Listener, and what thread pool handles the Logger component? A. BLOCKING IO, UBER B. UBER, Dedicated NIO Selector Pool C. CPU_LITE, CPU INTENSIVE D. Shared NIO Selector Pool, CPU_LITE	D	https://medium.com/the-mule-blog/mulesoft-core-execution-engine-and-processing-model-18955d6f5da6

12	Refer to the exhibit.  A customer is running Mule applications on Runtime Fabric for Self-Managed Kubernetes (RTF-BYOK8) in a multi-cloud environment. Based on this configuration, how do Agents and Runtime Manager communicate, and what is exchanged between them? A. Runtime Manager initiates traffic to Agents to check the status of clusters and auto-deploys new images when updates to Mule applications are available B. Agents communicate with Runtime Manager to provide updates on cluster status, download images, and update Mule applications (automatically when available) from the container registry C. Agents communicate with Runtime Manager to provide updates on cluster status, download images, and update Mule applications (when available and chosen to do so by customer) from the container registry D. Runtime Manager communicates with Agents to provide updates on cluster status, download images, and update Mule applications (when available and chosen to do so by customer) from the container registry	C
13	A Mule application contains a Batch Job scope with several Batch Step scopes. The Batch Job scope is configured with a batch block size of 25. A payload with 4,000 records is received by the Batch Job scope. When there are no errors, how does the Batch Job scope process records within and between the Batch Step scopes? A. The Batch Job scope processes multiple record blocks in parallel. Each Batch Step scope is invoked with a batch of 25 records in the payload of the received Mule event. For each Batch Step scope, all 4,000 records are processed in parallel. Individual records can jump ahead to the next Batch Step scope before the rest of the records finish processing in the current Batch Step scope. B. The Batch Job scope processes multiple record blocks in parallel, and a block of 25 records can jump ahead to the next Batch Step scope over an earlier block of records. Each Batch Step scope is invoked with one record in the payload of the received Mule event. For each Batch Step scope, all 25 records within a block are processed in parallel. All the records in a block must be completed before the block of 25 records is available to the next Batch Step scope. C. The Batch Job scope processes multiple record blocks in parallel, and a block of 25 records can jump ahead to the next Batch Step scope over an earlier block of records. Each Batch Step scope is invoked with one record in the payload of the received Mule event. For each Batch Step scope, all 25 records within a block are processed sequentially. All the records in a block must be completed before the block of 25 records is available to the next Batch Step scope. D. The Batch Job scope processes each record block sequentially. Each Batch Step scope is invoked with one record in the payload of the received Mule event. For each Batch Step scope, all 25 records within a block are processed sequentially. All 4,000 records must be completed before the blocks of records are available to the next Batch Step scope.	C
14	A stock trading company handles millions of trades a day and requires excellent performance and reliability within its stock trading system. The company operates a number of event-driven APIs implemented as Mule applications that are hosted on various customer-hosted Mule clusters and needs to enable message exchanges between the APIs within their internal network using shared message queues. What is an effective way to meet the cross-cluster messaging requirements of its event-driven APIs? A. Non-transactional JMS operations with a reliability pattern and manual acknowledgements B. Persistent VM queues with automatic acknowledgements C. JMS transactions with automatic acknowledgements D. extended Architecture (XA) transactions and XA connected components with manual acknowledgements	C
15	Why would an Enterprise Architect use a single enterprise-wide canonical data model (CDM) when designing an integration solution using Anypoint Platform? A. To reduce dependencies when integrating multiple systems that use different data formats B. To automate AI-enabled API implementation generation based on normalized backend databases from separate vendors C. To leverage a data abstraction layer that shields existing Mule applications from non-backward compatible changes to the model's data structure D. To remove the need to perform data transformation when processing message payloads in Mule applications	A
16	A cluster with two customer-hosted Mule runtimes is hosting an application that has a flow with a JMS listener configured to subscribe to messages from a topic destination. How should the JMS listener's primaryNodeOnly parameter and subscription parameter be configured so the JMS listener can receive messages in all the nodes of the cluster but only process each message once? A. Use the parameter primaryNodeOnly="false" on the JMS listener with a non-shared subscription B. Use the parameter primaryNodeOnly="false" on the JMS listener with a shared subscription C. Use the parameter primaryNodeOnly="true" on the JMS listener with a non-shared subscription D. Use the parameter primaryNodeOnly="true" on the JMS listener with a shared subscription	primaryNodeOnly="false": Setting this parameter to "false" ensures that the JMS listener is active on all nodes in the cluster, not just the primary node. This setting allows multiple instances of the JMS listener to run concurrently across different nodes, enabling them to consume messages from the JMS queue. https://docs.mulesoft.com/jms-connector/latest/jms-performance B Shared Subscription: Using a shared subscription means that all nodes will share the consumption of messages from the queue. This approach prevents duplicate message processing, as each message is delivered to only one listener instance within the cluster. This configuration ensures that message processing is balanced across the nodes, improving throughput and reliability. https://docs.mulesoft.com/jms-connector/latest/jms-topic-subscription
17	An organization is successfully using API-led connectivity. However, as the application network grows, all the manually performed tasks to publish, share and discover, register, apply policies to, and deploy an API are becoming repetitive, which is driving the organization to automate this process using efficient CI/CD pipelines. Considering Anypoint Platform's capabilities, how should the organization automate its API lifecycle? A. Use Anypoint CLI or Anypoint Platform REST APIs with a scripting language such as Groovy B. Use Maven with a custom configuration required for the API lifecycle C. Use Exchange REST APIs for API management and Maven for API deployment D. Use Runtime Manager REST APIs for API management and Maven for API deployment	A

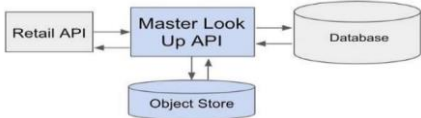
18	A payment processing company has implemented a Payment Processing API Mule application to process credit card and debit card transactions. Because the Payment Processing API handles highly sensitive information, the payment processing company requires that data must be encrypted both in-transit and at-rest. To meet these security requirements, consumers of the Payment Processing API must create request message payloads in a JSON format specified by the API, and the message payload values must be encrypted. How can the Payment Processing API validate requests received from API consumers? A. A Transport Layer Security (TLS) - Inbound policy can be applied in API Manager to decrypt the message payload and the Mule application implementation can then use the JSON Validation module to validate the JSON data B. The Mule application implementation can use the APIkit module to decrypt and then validate the JSON data C. The Mule application implementation can use the Validation module to decrypt and then validate the JSON data D. The Mule application implementation can use DataWeave to decrypt the message payload and then use the JSON Scheme Validation module to validate the JSON data	A	https://docs.mulesoft.com/gateway/latest/policies-included-tls https://docs.mulesoft.com/json-module/latest/json-schema-validation
19	An organization plans to migrate its deployment environment from an on-premises cluster to a Runtime Fabric(RTF) cluster. The on-premises Mule applications are currently configured with persistent object stores. There is a requirement to enable Mule applications deployed to the RTF cluster to store and share data across application replicas and through restarts of the entire RTF cluster. How can these reliability requirements be met? A. Replace persistent object stores with persistent VM queues in each Mule application deployment B. Install the Object Store pod on one of the cluster nodes C. Configure Anypoint Object Store v2 to share data between replicas in the RTF cluster D. Configure the Persistence Gateway in the RTF installation	D	https://docs.mulesoft.com/runtime-fabric/latest/persistence-gateway
20	A Mule solution implements an Experience API and System API as Mule applications that process messages in this order: 1. The Experience API receives a request Anypoint MQ message (REQU) with a payload containing a variable-length list of request objects. 2. The Experience API uses a For Each scope to split the list into individual objects and sends each object as a REQU_OBJECT message to an Anypoint MQ queue named REQUEST. 3. The System API listens on the REQUEST queue, processes each message Independently of all other messages, and sends a response message to a response queue named RESPONSE_OBJECT. The System API processing times can vary and are occasionally fairly long. 4. The Experience API listens on the RESPONSE_OBJECT queue and receives, transforms, and accumulates each response. 5. After receiving all the responses corresponding to the original REQU message, the Experience API creates and publishes a response Anypoint MQ message (RESP) to an Anypoint MQ queue named RESPONSE with a payload containing the list of responses returned by the System API, organized in the same order as the request objects originally sent in the REQU message. Assume successful response messages are returned by Experience API to the RESPONSE queue for all REQU messages. What is required so the Experience API can ensure the length and order of the list of objects in RESP and REQU match, while at the same time maximizing message throughput? A. Use a Scatter-Gather within the For Each scope to ensure response message order. Configure the Scatter-Gather with a persistent object store. B. Perform all communication involving the System API synchronously from within the For Each scope so that responses in RESPONSE_OBJECT are in the exact same order as request objects in the RESP message. Within the For Each scope, use an Async scope configured with a persistent object store. C. Use a persistent object store to keep track of the list length and all object indexes in the REQU message, both in the For Each scope and in all communication involving the System API. Set the maxConcurrency parameter for the receiving flow in the Experience API to a value greater than 1. D. After the For Each scope, add a second For Each scope configured with a persistent object store to collect response messages in the order in which they arrive, and then send the RESP message using this list of responses.	D	
21	A Mule application is required to periodically process a large data set from a back-end database to Salesforce CRM using a Batch-Job scope configured properly to process the high rate of records. The application is deployed to two CloudHub 1.0 workers the persistent VM queues option disabled(unchecked). What is the consequence if the worker crashes during records processing? A. Remaining records will be lost, resulting in unprocessed records B. The entire batch job will be processed from scratch by the second worker, resulting in some duplicate processing C. All the remaining records will be processed by a new replacement worker, resulting in "exactly once" processing of every record D. About half the remaining records will be processed by the second worker and the rest will be lost, resulting in some unprocessed records	A	https://docs.mulesoft.com/cloudhub/cloudhub-fabric#persistent-queues
22	A new Mule application has been deployed through Runtime Manager to CloudHub 1.0 using a CI/CD pipeline with sensitive properties set as cleartext. The Runtime Manager Administrator opened a high priority incident ticket about this violation of their security requirements indicating these sensitive properties values must not be stored or visible in Runtime Manager but should be changeable in Runtime Manager by Administrators with proper permissions. How can the Mule application be deployed while safely hiding the sensitive properties? A. Add an ArrayList of all the sensitive properties' names in the mule-artifact.json file of the application B. Add encrypted versions of the sensitive properties as global configuration properties in the Mule application C. Add a new wrapper.java.additional.XX parameter for each sensitive property in the wrapper.conf file used by the CI/CD pipeline scripts D. Create a variable for each sensitive property and declare them as hidden in the CI/CD pipeline scripts	B	https://docs.mulesoft.com/mule-runtime/latest/secure-configuration-properties
23	A Mule application uses an HTTP Request operation to invoke an external API. The external API follows the HTTP specification for proper status code usage. What is a possible cause when a 3xx status code is returned to the HTTP Request operation from the external API? A. The request was received and processed by the external API B. The request was redirected to a different URL by the external API C. The request was received and rejected by the external API D. The request was not received by the external API	B	https://medium.com/@amrutha.tamada/when-you-are-working-with-an-application-especially-a-mule-application-and-suddenly-an-error-a084df08e501
24	A Mule application is synchronizing customer data between two different database systems. What is a main benefit of using extended Architecture (XA) transactions rather than local transactions to synchronize these two different database systems? A. An XA transaction handles the largest number of requests in the shortest time B. An XA transaction synchronizes the database systems with the least amount of Mule configuration or coding C. An XA transaction automatically rolls back operations against both database systems if any operation fails D. An XA transaction writes to both database systems quickly and simultaneously	C	https://docs.mulesoft.com/mule-runtime/latest/xa-transactions

25	A company is designing an integration Mule application to process orders by submitting them to a back-end system for offline processing. Each order will be received by the Mule application through an HTTPS POST and must be acknowledged immediately. Once acknowledged, the order will be submitted to a back-end system. Orders that cannot be successfully submitted due to rejections from the back-end system will need to be processed manually(outside the back-end system). The Mule application will be deployed to a customer-hosted runtime and will be able to use an existing ActiveMQ broker if needed. The ActiveMQ broker is located inside the organization's firewall. The back-end system has a track record of unreliability due to both minor network connectivity issues and longer outages. What combination of Mule application components and ActiveMQ queues are required to ensure automatic submission of orders to the back-end system while supporting but minimizing manual order processing? A. A Batch Job scope to call the back-end system An Until Successful scope containing Object Store components for long retries A dead-letter object store configured in the Mule application B. One or more on Error scopes to assist calling the back-end system One or more ActiveMQ long-retry queues A persistent dead-letter Object Store configured in the CloudHub Object Store service C. An Until Successful scope to call the back-end system One or more ActiveMQ long-retry queues One or more ActiveMQ dead-letter queues for manual processing D. One or more On Error scopes to assist calling the back-end system An Until Successful scope containing VM components for long retries A persistent dead-letter VM queue configured in CloudHub	C
26	An organization has strict unit test requirements that mandate every Mule application must have an MUnit test suite with a test case defined for each flow and a minimum test coverage of 80 percent. A developer is building an MUnit test suite for a newly developed Mule application that sends API requests to an external REST API. What is an effective approach for successfully executing the MUnit tests of this Mule application while still achieving the required test coverage for the MUnit tests? A. Mock the REST API invocations in the MUnit tests, and then forward mocked requests to the external REST API B. Create a mocking service flow to simulate standard responses from the REST API, and then configure the Mule flows to call the mocking service flow C. Mock the REST API invocations in the Munit tests, and then return a mock response for those invocations D. Invoke the external endpoint of the REST API from the Mule flows	C
27	A company is designing a Mule application to consume batch data from a partner's FTPS server, ftps.partner.com. The data files have been compressed (but not encrypted) and then digitally signed by the partner using Pretty Good Privacy (PGP). What inputs are required for the application to securely login to the FTPS server and to consume these files? A. The company's FTPS server login username and password A TLS context trust store containing a public certificate for the company The company's PGP public key that was used to sign the files B. The partner's PGP public key used by the company to login to the FTPS server A TLS context key store containing the private key for the company The partner's PGP private key that was used to sign the files C. The company's FTPS server login username and password A TLS context trust store containing a public certificate for ftps.partner.com The partner's PGP public key that was used to sign the files D. The partner's PGP public key used by the company to login to the FTPS server A TLS context key store containing the private key for ftps.partner.com The company's PGP private key that was used to sign the files	C
28	An organization is struggling with frequent plugin version upgrades and external plugin project dependencies. The team wants to minimize the Impact on applications by creating best practices that will define a set of default dependencies across all new and in-progress projects. How can these best practices be achieved with the applications having the least amount of responsibility? A. Create a Mule domain project with all the dependencies defined in its pom.xml file and add each application to the domain project B. Create a Mule plugin project with all the dependencies and add it as a dependency in each application's pom.xml file C. Add all the dependencies in each application's pom.xml file D. Create a parent POM with all the required dependencies and reference it in each application's pom.xml file	D
29	Refer to the exhibit. One of the back-end systems invoked by the API implementation enforces rate limits on the number of requests a particular client can make. Both the back-end system and the API implementation are deployed to several non- production environments, including the staging environment, and to a production environment. Rate limiting of the back-end system applies to all non-production environments. The production environment, however, does not have any rate limiting. What is the most effective approach to conduct performance tests of the API implementation in the non-production staging environment?  A. Use MUnit to simulate standard responses from the back-end system. Then, conduct performance tests to identify other bottlenecks in the system. B. Create a mocking service that replicates the back-end system's production performance characteristics. Then, configure the API implementation to use the mocking service and conduct the performance tests. C. Conduct scaled-down performance tests in the staging environment against the rate-limited back-end system. Then, upscale performance results to full production scale. D. Include logic within the API implementation that bypasses invocations of the back-end system in the staging environment, and invoke a mocking service that replicates typical back-end system responses. Then, conduct performance tests using this API implementation.	B

30	An organization's security requirements mandate centralized control at all times over authentication and authorization of external applications when invoking web APIs managed on Anypoint Platform. What Anypoint Platform feature is the most straightforward and maintainable to use to meet this requirement? A. Client management configured in Access Management B. Enterprise Security module coded in Mule applications C. External access configured in API Manager D. Identity management configured in Access Management	A	https://docs.mulesoft.com/access-management/managing-api-clients
31	An organization uses a four-node customer-hosted Mule runtime cluster to host one stateless API implementation. The API is accessed over HTTPS through a load balancer that uses round-robin for load distribution. Each node in the cluster has been sized to accept four times the current number of requests. Two nodes in the cluster experience a power outage and are no longer available. The load balancer detects the outage and blocks the two unavailable nodes from receiving further HTTP requests. What performance-related consequence is guaranteed to happen, on average, assuming the remaining cluster nodes are fully operational? A. 100% increase in the average response time of the API implementation B. 50% reduction in the throughput of the API implementation C. 50% increase in the number of thread pools on the remaining servers D. 100% increase in the number of requests received by each remaining node	C	
32	What requirement prevents using Anypoint MQ as the messaging broker for a Mule application? A. When the payload sent through the message broker must use XML format B. When the payload sent through the message broker must be encrypted C. When the messaging broker must support point-to-point messaging D. When the messaging broker must be deployed on-premises	D	
33	An organization plans to use the Anypoint Platform audit logging service to log Anypoint MQ actions. What consideration must be kept in mind when leveraging Anypoint MQ Audit Logs? A. Anypoint MQ Audit Logs include logs for sending, receiving, or browsing messages B. Anypoint MQ Audit Logs include logs for failed Anypoint MQ operations C. Anypoint MQ Audit Logs include logs for queue create, delete, modify, and purge operations D. Viewing Anypoint MQ Audit Logs requires having the Organization Administrator role	D	https://docs.mulesoft.com/mq/mq-audit-logs
34	A retail company is implementing a MuleSoft API to get inventory details from two vendors by invoking each vendor's online applications. Due to network issues, the invocations to the vendor applications are timing out intermittently, but the requests are successful after re-invoking each vendor application. What is the most performant way of implementing the API to invoke each vendor application and to retry invocations that generate timeout errors? A. Use a For-Each scope to invoke the two vendor applications in series, one after the other. Place the For-Each scope inside an Until-Successful scope to retry requests that raise timeout errors. B. Use a Choice scope to invoke each vendor application on a separate route. Place the Choice scope inside an Until-Successful scope to retry requests that raise timeout errors. C. Use a Scatter-Gather scope to invoke each vendor application on a separate route. Use an until-Successful scope in each route to retry requests that raise timeout errors. D. Use a Round-Robin scope to invoke each vendor application on a separate route. Use a Try-Catch scope in each route to retry requests that raise timeout errors.	C	
35	An organization is using MuleSoft to develop APIs and automatically deploy them to both CloudHub and on-premises targets. Recently, it has decided to include Runtime Fabric deployment targets as well, and Mule infrastructure is set up for this option. What can now be used to automate deploying Mule applications to all three types of deployment targets? A. The Mule Maven plugin B. The Runtime Manager agent configuration file C. The Object Store v2 REST API D. Anypoint Monitoring agent configuration file	A	https://docs.mulesoft.com/mule-runtime/4.4/mmp-concept
36	When a Mule application using VM queues is deployed to a customer-hosted cluster or multiple CloudHub v1.0 workers/replicas, how are messages consumed across the nodes? A. Sequentially, from a dedicated Anypoint MQ queue B. Sequentially, only from the primary node C. In a non-deterministic way D. Round-robin, within an XA transaction	C	
37	A healthcare client must connect its Mule APIs that are deployed to CloudHub 1.0 to systems in an existing on-premises datacenter through private and secure connections. Which network architecture supports these security requirements to connect the Mule APIs to the on-premises datacenter? A. An Anypoint VPC connected to the on-premises network using a dedicated load balancer Matching IP includelisting in the load balancer and firewall rules in the VPC and on-premises network B. An Anypoint VPC connected to the on-premises network using an IPsec tunnel Matching firewall rules in the VPC and on-premises network C. Static IP addresses for the Mule applications deployed to the CloudHub 1.0 Shared Worker Cloud Matching firewall rules and IP includelisting in the on-premises network D. Static IP addresses for the Mule applications deployed to the CloudHub 1.0 Shared Worker Cloud Matching IP includelisting in the load balancer and firewall rules in the VPC and on- premises network	B	

38	Following MuleSoft's recommended best practices for API governance and API policies, a project team has used RAML specifications to document and publish functional requirements and detailed design definitions of its APIs. These API specifications have been used by various stakeholders to implement APIS. Later, the project team requires all API specifications to be augmented with an additional non-functional requirement (NFR) to protect the backend services from a high rate of requests, according to defined service-level agreements (SLAs). The NFR'S SLAs are based on a new tiered subscription level "Gold", "Silver", or "Platinum" that must be tied to a new parameter that is being added to the Accounts object in their enterprise data model. Following MuleSoft's recommended best practices, how should the project team now convey the necessary non-functional requirement to stakeholders? A. Create and deploy API proxies in API Manager for the NFR, change the baseUrl in each API specification to the corresponding API proxy implementation endpoint, and publish each modified API specification to Exchange B. Update each API specification with comments about the NFR'S SLAS and publish each modified API specification to Exchange C. Update each API specification with a shared RAML fragment required to implement the NFR and publish the RAML fragment and each modified API specification to Exchange D. Create a shared RAML fragment required to implement the NFR, list each API implementation endpoint in the RAML fragment, and publish the RAML fragment to Exchange	C	
39	A leading eCommerce giant will use MuleSoft APIs on Runtime Fabric (RTF) to process customer orders. Some customer-sensitive information, such as credit card information, is required in request payloads or is included in response payloads in some of the APIs. Other API requests and responses are not authorized to access some of this customer-sensitive information but have been implemented to validate and transform based on the structure and format of this customer-sensitive information (such as account IDs, phone numbers, and postal codes). What approach configures an API gateway to hide sensitive data exchanged between API consumers and API implementations, but can convert tokenized fields back to their original value for other API requests or responses, without having to recode the API implementations? A. Create both masking and tokenization formats and use both to apply a tokenization policy in an API gateway to mask sensitive values in message payloads with **** characters, and apply a corresponding detokenization policy to return the original values to other APIs B. Create a masking format and use it to apply a tokenization policy in an API gateway to mask sensitive values in message payloads with **** characters, and apply a corresponding detokenization policy to return the original values to other APIs C. Use a field-level encryption policy in an API gateway to replace sensitive fields in message payload with encrypted values, and apply a corresponding field-level decryption policy to return the original values to other APIS D. Create a tokenization format and use it to apply a tokenization policy in an API gateway to replace sensitive fields in message payload with similarly formatted tokenized values, and apply a corresponding detokenization policy to return the original values to other APIS	D	https://docs.mulesoft.com/anypoint-security/apply-tokenization-policy-to-api
40	An external API frequently invokes an Employees System API to fetch employee data from a MySQL database. The architect must design a caching strategy to query the database only when there is an update to the Employees table or else return a cached response in order to minimize the number of redundant transactions being handled by the database. What must the architect do to achieve the caching objective? A. Use an On Table Row operation configured with the Employees table, call invalidate cache, and hardcode the new Employees data to cache. Use an object-store-caching-strategy and set the expiration interval to 1 hour. B. Use an On Table Row operation configured with the Employees table and call invalidate cache. Use an object-store-caching-strategy and the default expiration interval. C. Use a Scheduler with a fixed frequency set to every hour to trigger an invalidate cache flow. Use an object-store-caching-strategy and the default expiration interval. D. Use a Scheduler with a fixed frequency set to every hour, triggering an invalidate cache flow. Use an object-store-caching-strategy and set the expiration interval to 1 hour.	B	
41	An airline is architecting an API-led connectivity project to integrate its flight data into an online aggregation website. The connections must allow for secured communication, high performance, and asynchronous message exchange. What are suitable interface technologies for the connectors in this integration assuming that MuleSoft fully supports those technologies and that Anypoint Connectors exist for these interfaces? A. XML over ActiveMQ XML over FTP XML/REST over HTTPS B. CSV over FTP YAML over TLS JSON over HTTPS C. AsyncAPI over SQS AMQP with RabbitMQ JSON/REST over HTTPS D. SOAP over HTTPS IIOP over TLS gRPC over HTTPS	C	
42	An organization has built a large monolithic application over the years and is currently looking to transition to a microservices architecture. During this transition phase, it has built some System APIs as Mule applications that provide access to different pieces of functionality of the monolithic application. The System APIs are deployed to Mule runtimes in a customer- hosted EC2 instance in AWS. The System APIs communicate directly with one database used by the monolithic application. Currently, a Database connector and configuration for that connector must be built into every project. There is a plan to partition this database into various smaller data stores in the future. What is the MuleSoft-recommended best practice to share the connector and configuration information among the APIs? A. Build a Mule domain project, add the Database connector and configuration to it, and reference this one domain project from each System API B. Build a separate Mule domain project for each API, and configure each of them to use a file on a shared file store to load the configuration information dynamically C. Build another System API that connects to the database, and refactor all the other APIs to make requests through the new System API to access the database D. Create an API proxy for each System API and share the Database connector configuration with all the API proxies via an automated policy	A	

43	An organization has deployed Runtime Fabric on a single server configured with an eight-node cluster using the performance profile. An API uses a non-persistent Object Store for maintaining state data from deployed Mule applications. What will be the impact to the state data if the server crashes? A. State data is lost B. State data is preserved C. State data is duplicated D. State data is rolled back to a previously saved version	A	https://help.salesforce.com/s/articleView?id=001123185&type=1
44	An organization is in the process of building automated deployments using a CI/CD process. As a part of automated deployments, it wants to apply policies to API instances. What tool can the organization use to promote and deploy API Manager policies? A. Anypoint CLI B. MUnit Maven plugin C. Mule Maven plugin D. Runtime Manager agent	A	https://www.mulesoft.com/resources/api/continuous-integration-continuous-delivery
45	An application deployed in a CloudHub production environment is throwing intermittent exceptions. To troubleshoot the issue, a developer wants to enable DEBUG level logging in production for the org.apache.cxf package in the deployed Mule application. How should the developer update the logging configuration in order to enable this package specific debugging? A. In Anypoint Monitoring, define a logging search query with class property set to org.apache.cxf and level set to DEBUG B. In the Mule application's log4j2.xml file, add an AsyncLogger element with name property set to org.apache.cxf and level set to DEBUG, then redeploy the Mule application in the CloudHub production environment C. In the Mule application's log4j2.xml file, change the root logger's level property to DEBUG, then redeploy the Mule application to the CloudHub production environment D. In Anypoint Runtime Manager, in the Deployed Application Properties tab for the Mule application, add a line item with DEBUG level for package org.apache.cxf and apply the changes	B	https://help.salesforce.com/s/articleView?id=001114432&type=1 https://docs.mulesoft.com/mule-runtime/3.9/logging-in-mule
46	Which Salesforce API is invoked to deploy, retrieve, create, update, or delete customization information, such as custom object definitions using Mule Salesforce Connectors in a Mule application? A. sObject Platform Action API B. User Interface API C. Metadata API D. Process Rules API	C	
47	A corporation has deployed Mule applications to different customer-hosted Mule runtimes. Mule applications deployed to these Mule runtimes are managed by Anypoint Platform. What needs to be installed or configured (if anything) to monitor these Mule applications from Anypoint Monitoring, and how is monitoring data from each Mule application sent to Anypoint Monitoring? A. Enable monitoring of individual Mule applications from the Runtime Manager application settings. Runtime Manager sends monitoring data to Anypoint Monitoring for each deployed Mule application. B. Install a Runtime Manager agent on each Mule runtime. Each Runtime Manager agent sends monitoring data from the Mule applications running in its Mule runtime to Runtime Manager, then Runtime Manager sends monitoring data to Anypoint Monitoring. C. Leave the out-of-the-box Anypoint Monitoring agent unchanged in its default Mule runtime installation. Each Anypoint Monitoring agent sends monitoring data from the Mule applications running in its Mule runtime to Runtime Manager, then Runtime Manager sends monitoring data to Anypoint Monitoring. D. Install an Anypoint Monitoring agent on each Mule runtime. Each Anypoint Monitoring agent sends monitoring data from the Mule applications running in its Mule runtime to Anypoint Monitoring.	D	https://docs.mulesoft.com/monitoring/am-installing
48	An insurance company is designing a hybrid, load-balanced, single-cluster runtime production environment. Due to performance Service Level Agreement (SLA) goals, it is looking into running the Mule applications in an active-active multi-node cluster configuration. What should be considered when running its Mule applications in this type of environment? A. A Mule application deployed to multiple nodes runs in isolation from the other nodes in the cluster B. Although the cluster environment is fully installed, configured, and running, it will not process any requests until an outage condition is detected by the primary node in the cluster C. All event sources, regardless of type, can be configured as the target source by the primary node in the cluster D. An external load balancer is required to distribute incoming TCP/HTTP requests throughout the cluster nodes	D	https://docs.mulesoft.com/mule-runtime/latest/mule-high-availability-ha-clusters
49	How does a timeout attribute in a configured transaction manager help inform design decisions while using a JMS connector listening for incoming messages in an eXtended Architecture(XA) transaction? A. After the timeout is exceeded, stale JMS Consumer threads are destroyed and threads are created B. The timeout defines the time that is allowed to pass without the transaction ending explicitly, and after the timeout expires, the transaction rolls back C. The timeout specifies the allowed to pass between receiving JMS messages on the same JMS connection, and then after the timeout, a new JMS connection is established D. The time allowed to pass between committing the transaction and the completion of the Mule flow, and then after the timeout, flow processing triggers an error	B	https://docs.mulesoft.com/mule-runtime/3.9/xa-transactions
50	An organization plans to extend its Mule APIs to the EU (Frankfurt) region. Currently, all Mule applications are deployed to CloudHub 1.0 in the default North American region, from the North America control plane, following this naming convention: {API-name}-{environment} (for example, Orders- sapi-dev, Orders-sapi-qa, Orders-sapi-prod, etc.). There is no network restriction to block communications between APIs. What strategy should be implemented in order to deploy the same Mule APIs to the CloudHub 1.0 EU region from the North America control plane, as well as to minimize latency between APIs and target users and systems in Europe? A. In Runtime Manager, for each Mule application deployment, set the Region property to EU (Frankfurt) and reuse the same Mule application name as in the North American region. Communicate the new urls {API-name}-{environment}.de-c1.cloudhub.io to the consuming API clients in Europe. B. In API Manager, set the Region property to EU (Frankfurt) to create an API proxy named {API-name}-proxy-{environment} for each Mule application. Communicate the new url {API-name}-proxy-(environment).de-c1.cloudhub.io to the consuming API clients in Europe. C. In Runtime Manager, for each Mule application deployment, leave the Region property blank (default) and change the Mule application name to {API-name}- (environment).de-c1. Communicate the new urls {API-name}-(environment).de-c1.cloudhub.io to the consuming API clients in Europe. D. In API Manager, leave the Region property blank (default) to deploy an API proxy named {API-name}-proxy-(environment).de-c1 for each Mule application. Communicate the new url {API-name}-proxy-(environment).de-c1.cloudhub.io to the consuming API clients in Europe.	A	

51	In a Mule 4 application that will be deployed to customer-hosted Mule runtimes, the Java module's Invoke Static operation needs to invoke a static method in a dependency JAR file. How can the dependency be made visible to the Java module's classloader? A. In the Java Invoke Static module configuration, configure the path and name of the dependency JAR file B. Use a Maven command to include the dependency JAR file when packaging the application C. Configure the dependency as a shared library in the project's POM file D. Add the dependency JAR file to the Java classpath by setting JVM parameters	C	https://docs.mulesoft.com/mule-runtime/latest/about-classloading-isolation#shared-libraries
52	Which operation can be performed through a JMX agent enabled in a Mule application? A. View object store entries B. Set a particular log4j2 log level to TRACE C. Deploy a Mule application D. Replay an unsuccessful message	B	https://docs.mulesoft.com/mule-runtime/3.9/jmx-management
53	An organization plans to migrate all its Mule applications to Runtime Fabric(RTF). Currently, all Mule applications have been deployed to CloudHub using automated CI/CD scripts. Which steps should be taken to properly migrate the applications from CloudHub to RTF, keeping the same deployment strategy? A. No changes need to be made to the pom.xml and CI/CD scripts should be modified as per the RTF configurations B. A <i>runtimeFabric</i> dependency should be added as mule-plugin to pom.xml and CI/CD scripts should be modified as per the RTF configurations C. A <i>runtimeFabricDeployment</i> profile should be added to pom.xml in all the Mule applications and CI/CD scripts should be modified as per the RTF configurations D. A <i>runtimeFabricDeployment</i> profile should be added to Mule configuration files in all the Mule applications and CI/CD scripts should be modified as per the RTF configurations	C	https://docs.mulesoft.com/mule-runtime/latest/deploy-to-rtf
54	A manufacturing company plans to deploy Mule applications to its own Azure Kubernetes service infrastructure. The organization wants to make the Mule applications more available and robust by deploying each Mule application to an isolated Mule runtime in a Docker container while managing all the Mule applications from the MuleSoft-hosted control plane. What choice of runtime plane meets these organizational requirements? A. Anypoint Service Mesh B. Customer-hosted self-provisioned runtime plane C. Anypoint Runtime Fabric D. CloudHub 2.0	C	https://docs.mulesoft.com/runtime-fabric/latest/
55	An Order microservice and a Fulfillment microservice are being designed to communicate with their clients through message-based integration (and NOT through API invocations). The Order microservice publishes an Order message (a kind of command message) containing the details of an order to be filled. The intent is that Order messages are only consumed by one Mule application: the Fulfillment microservice. The Fulfillment microservice consumes Order messages, fills the orders described there in, and then publishes a Fulfillment message (a kind of event message). Each Fulfillment message can be consumed by any Interested Mule application, and the Order microservice is one such Mule application. What is the recommended choice of message broker(s) and message destination(s) in this scenario? A. Order messages are sent to a JMS queue. Fulfillment messages are sent to a JMS topic. Both microservices Interact with the same JMS provider (message broker) Instance, which must therefore scale to support the load of both microservices B. Order messages are sent to an Anypoint MQ exchange Fulfilled messages are sent directly to an Anypoint MQ queue Both microservice interacts with Anypoint MQ as message broker, which must therefore scale to support the load of both microservice. C. Order messages are sent to a JMS topic Fulfillment messages are sent to a JMS queue The Order microservice Interacts with one JMS provider (message broker) and the Fulfillment microservice Interacts with a different JMS provider so that both message brokers can be chosen and scaled to best support the load of each microservice D. Order messages are sent directly to the Fulfillment microservices Fulfilled messages are sent directly to the Order microservice. The Order microservice interacts with one Advanced Message Queuing Protocol(AMQP)-compatible message broker and the Fulfillment microservice Interacts with a different AMQP-compatible message broker, so that both message brokers can be chosen and scaled to best support redelivery to each microservice and future auditing and intergrations	A	
56	An organization is designing multiple Mule applications to run on CloudHub in a single Anypoint VPC and that must share data using a common persistent Anypoint Object Store v2 (OSv2). Which design gives these Mule applications access to the same Object Store instance? A. A VM connector configured to directly access the persistence queue of the persistent Object Store B. An Anypoint MQ connector configured to directly access the persistent Object Store C. The Object Store v2 can be shared across CloudHub applications with a configured OS v2 connector D. The Object Store v2 REST API configured to access the persistent Object Store	D	https://docs.mulesoft.com/object-store/ https://docs.mulesoft.com/object-store/osv2-guide
57	Refer to the exhibit.  <pre> graph LR RetailAPI[Retail API] <--> MLU[Master Look Up API] MLU <--> Database[(Database)] MLU <--> ObjectStore[(Object Store)] </pre> A manufacturing company is developing a new set of APIs for its retail business. One of the APIs is a Master Look Up API, which is a System API. The API uses a persistent object-store. This API will be used by almost all other APIs to provide master lookup data. The Master Look Up API is deployed on two CloudHub workers of 0.1 vCore each because there is a lot of master data to be cached. Most of the master lookup data is stored as a key-value pair. The cache gets refreshed if the key is not found in the cache. During performance testing, it was determined that the Master Look Up API has a high response time due to the latency of database queries executed to fetch the master lookup data. What two methods can be used to resolve these performance issues? A. Implement HTTP caching policy for all GET endpoints in the Master Look Up API. B. Upgrade vCore size from 0.1 vCore to 0.2 vCore. C. Implement HTTP caching policy for all GET endpoints for the Master Look Up API. D. Implement locking to synchronize access to Object Store.	C D	

58	A Mule application is designed to call the Google Maps API to perform a distance computation, and is deployed to CloudHub. At a minimum, what (if anything) should be configured in the TLS context of the HTTP Request configuration to meet these requirements? A. Request a private key from Google, create a PKCS# 12 file with it, and add it in KeyStore as part of the TLS context B. Download the Google public certificate from a browser, generate a JKS file from it, and add it in KeyStore as part of the TLS context C. The configuration is built-in and nothing extra is required for the TLS context D. Download the Google public certificate from a browser, generate a JKS file from it, and add it in TrustStore as part of the TLS context	D	https://docs.mulesoft.com/mule-runtime/latest/tls-configuration
59	A project team is working on an API implementation using an API RAML specification as a starting point. The team has updated the RAML specification to include new operations, and has published a new version to Exchange. Meanwhile, another team is working on an API client Mule application to consume the same API implementation. During this development, what must be performed by the Mule application team to take advantage of the newly added operations? A. Scaffold the API client application with APIkit using the new RAML specification B. Update the API connector in the API implementation and publish to Exchange C. Update the REST connector from Exchange in the API client application D. Copy the API implementation's updated APIkit component into the API client	C	
60	A marketing organization is designing a Mule application to process campaign data. The Mule application will periodically check for a file in a SFTP location and process the records in the file. The size of the file can vary from 10MB to 5GB. Due to the limited availability of vCores, the Mule application is deployed to a single CloudHub worker configured with vCore size 0.2. The application must transform and send different formats of this file to three different downstream SFTP locations. What is the most idiomatic (used for its intended purpose) and performant way to configure the SFTP operations or event sources to process the large files to support these deployment requirements? A. Use an in-memory non-repeatable stream B. Use a file-stored non-repeatable stream C. Use an in-memory repeatable stream D. Use a file-stored repeatable stream	D	https://docs.mulesoft.com/mule-runtime/4.4/streaming-about
61	A corporation has deployed multiple Mule applications implementing various public and private APIs to different CloudHub workers. These APIs are critical applications that must be highly available and inline with the reliability SLA as defined by stakeholders. How can API availability (liveness or readiness) be monitored so the Ops team receives outage notifications? A. Enable monitoring of individual applications from Anypoint Monitoring B. Configure alerts with failure conditions in Runtime Manager C. Configure alerts with failure conditions in API Manager D. Use Anypoint Functional Monitoring to test APIs' functional behavior	D	https://docs.mulesoft.com/monitoring/ https://docs.mulesoft.com/api-functional-monitoring/afm-in-anypoint-platform
62	A Mule application must periodically process a large data set, which varies from 6GB-8GB, from a back-end database and write transformed data to an FTPS server using a property configured batch job scope. The performance requirements of the application are approved to run in a CloudHub 0.2 vCore with 8GB storage capacity, and the concurrency requirements are met. How can the high rate of records be effectively managed this application? A. Use streaming with a file store repeatable strategy for reading records from the database, and batch aggregator with streaming to write to FTPS B. Use streaming with an in-memory repeatable store strategy for reading records from the database, and batch aggregator with streaming to write to FTPS C. Use streaming with a file store repeatable strategy for reading records from the database and batch aggregator with an optimal size D. Use streaming with a file store repeatable strategy for reading records from the database and batch aggregator without any required configuration	A	
63	ACME Tractors is designing a Mule application named Inventory that uses a persistent object store. The Inventory Mule application is deployed to CloudHub and is configured to use Object Store v2. Another Mule application named Cleanup is being developed to delete values from the Inventory Mule application's persistent object store. The Cleanup Mule application will also be deployed to CloudHub. What is the most direct way for the Cleanup Mule application to delete values from the Inventory Mule application's persistent object store with the least latency? A. Use a VM connector configured to directly access the persistent queue of the Inventory Mule application's persistent object store B. Use an Object Store connector configured to access the Inventory Mule application's persistent object store C. Use the Object Store v2 REST API configured to access the Inventory Mule application's persistent object store D. Use an Anypoint MQ connector configured to directly access the Inventory Mule application's persistent object store	C	https://docs.mulesoft.com/object-store/osv2-guide
64	A Mule application is running on a customer-hosted Mule runtime in an organization's network. The Mule application acts as a producer of asynchronous Mule events. Each Mule event must be broadcast to all interested external consumers outside the Mule application. The Mule events should be published in a way that is guaranteed in normal situations and also minimizes duplicate delivery in less frequent failure scenarios. The organizational firewall is configured to only allow outbound traffic on ports 80 and 443. Some external event consumers are within the organizational network, while others are located outside the firewall. What Anypoint Platform service is most idiomatic (used for its intended purpose) for publishing these Mule events to all external consumers while addressing the desired reliability goals? A. CloudHub VM queues B. CloudHub Shared Load Balancer C. Anypoint MQ D. Anypoint Exchange	C	https://docs.mulesoft.com/mq/

65	A Mule application is built to support a local transaction for a series of operations on a single database. The Mule application has a Scatter-Gather that participates in the local transaction. What is the behavior of the Scatter-Gather when running within this local transaction? A. Execution of all routes within the Scatter-Gather occurs in parallel Any error that occurs inside the Scatter-Gather will result in a rollback of all the database operations B. Execution of each route within the Scatter-Gather occurs sequentially Any error that occurs inside the Scatter-Gather will NOT result in a rollback of any of the database operations C. Execution of each route within the Scatter-Gather occurs in parallel Any error that occurs inside the Scatter-Gather will NOT result in a rollback of any of the database operations D. Execution of each route within the Scatter-Gather occurs sequentially Any error that occurs inside the Scatter-Gather will result in a rollback of all the database operations	D	
66	A Mule application is being designed for deployment to a single CloudHub worker. The Mule application will have a flow that connects to a SaaS system to perform some operations each time the flow is invoked. The SaaS system connector has operations that can be configured to request a short-lived token (fifteen minutes) that can be reused for subsequent connections within the fifteen minute time window. After the token expires, a new token must be requested and stored. What is the most performant and idiomatic (used for its intended purpose) Anypoint Platform component or service to use to support persisting and reusing tokens in the Mule application to help speed up reconnecting the Mule application to the SaaS application? A. Persistent object store B. Variable C. Database D. Non persistent object store	D	
67	To implement predictive maintenance on its machinery equipment, ACME Tractors has installed thousands of IoT sensors that will send data for each machinery asset as sequences of JMS messages, in near real-time, to a JMS queue named SENSOR_DATA on a JMS server. The Mule application contains a JMS Listener operation configured to receive incoming messages from the JMS server's SENSOR_DATA JMS queue. The Mule application persists each received JMS message, then sends a transformed version of the corresponding Mule event to the machinery equipment back-end systems. The Mule application will be deployed to a multi-node, customer-hosted Mule runtime cluster. Under normal conditions, each JMS message should be processed exactly once. How should the JMS Listener be configured to maximize performance and concurrent message processing of the JMS queue? A. Set numberOfConsumers to a value greater than one Set primaryNodeOnly = false B. Set numberOfConsumers = 1 Set primaryNodeOnly = true C. Set numberOfConsumers = 1 Set primaryNodeOnly = false D. Set numberOfConsumers to a value greater than one Set primaryNodeOnly = true	C	
68	The AnyAirline organization's passenger reservations center is designing an integration solution that combines invocations of three different System APIs (bookFlight, bookHotel, and bookCar) in a business transaction. Each System API makes calls to a single database. The entire business transaction must be rolled back when at least one of the APIs fails. What is the most idiomatic (used for its intended purpose) way to integrate these APIs in near real-time that provides the best balance of consistency, performance, and reliability? A. Implement local transactions within each API implementation Configure each API implementation to also participate in the same extended Architecture (XA) transaction Implement caching in each API implementation to improve performance B. Implement local transactions in each API implementation Coordinate between the API implementations using a Saga pattern Apply various compensating actions depending on where a failure occurs C. Implement extended Architecture (XA) transactions between the API implementations Coordinate between the API implementations using a Saga pattern Implement caching in each API implementation to improve performance D. Implement an extended Architecture (XA) transaction manager in a Mule application using a Saga pattern Connect each API implementation with the Mule application using XA transactions Apply various compensating actions depending on where a failure occurs	B	
69	ACME Tractors is creating a Mule application that will be deployed to CloudHub. The Mule application has a property named dbPassword that stores a database user's password. The organization's security standards indicate that the dbPassword property must be hidden from every Anypoint Platform user after the value is set in the Runtime Manager Properties tab. What configuration in the Mule application helps hide the dbPassword property value in Runtime Manager? A. Add the dbpassword property to the secureProperties section of the mule-artifact.json file B. Use secure::dbPassword as the property placeholder name and store the cleartext (unencrypted) value in a secure properties placeholder file C. Use secure::dbPassword as the property placeholder name and store the properly encrypted value in a secure properties placeholder file D. Add the dbpassword property to the secureProperties section of the pom.xml file	A	https://docs.mulesoft.com/cloudhub/secure-application-properties#create-safely-hidden-application-properties

70	An organization has previously provisioned its own AWS virtual private cloud (VPC) that contains several AWS instances. The organization now needs to use CloudHub to host a Mule application that will implement a REST API. Once deployed to CloudHub, this Mule application must be able to communicate securely with the customer-provisioned AWS VPC resources within the same region, without being interceptable on the public internet. What Anypoint Platform features should be used to meet these network communication requirements between CloudHub and the existing customer-provisioned AWS VPC? A. Configure a MuleSoft-hosted (CloudHub) dedicated load balancer with mapping rules that allow secure traffic from the range of IP addresses located in the customer-provisioned AWS VPC to access the Mule application B. Add a MuleSoft-hosted (CloudHub) Anypoint VPC configured with VPC peering to the range of IP addresses located in the customer-provisioned AWS VPC C. Add default API allowlist policies to API Manager to automatically secure traffic from the range of IP addresses located in the customer-provisioned AWS VPC to access the Mule application D. Configure an external identity provider (IdP) in Anypoint Platform with certificates from an AWS Transit Gateway for the customer-hosted AWS VPC, where the certificates allow the range of IP addresses located in the customer-provisioned AWS VPC	B	
71	An API is being implemented using the components of Anypoint Platform. The API implementation needs to be managed and governed (by applying API policies) on Anypoint Platform. What needs to be done in Anypoint Platform before the API implementation can be governed by Anypoint Platform? A. The API must be shared with the potential developers through an API portal so API consumers can interact with the API B. The API must be published to Anypoint Exchange and a corresponding API instance ID must be obtained from API Manager to be used in the API implementation C. The API implementation source code must be committed to a source control management system (such as GitHub) D. A RAML definition of the API must be created in API Designer so it can then be published to Anypoint Exchange	B	
72	ACME Tractors has implemented a continuous integration (CI) lifecycle that promotes Mule applications through code, build, and 4 test stages. To standardize the organization's CI journey, a new dependency control approach is being designed to store artifacts that include information such as dependencies, versioning, and build promotions. To implement these process improvements, the organization will now require developers to maintain all dependencies related to Mule application code in a shared location. What is the most idiomatic (used for its intended purpose) type of system the organization should use in a shared location to standardize all dependencies related to Mule application code? A. The Anypoint Object Store service at cloudhub.io B. A binary artifact repository C. A MuleSoft-managed repository at repository.mulesoft.org D. API Community Manager	B	
73	ACME Tractors is designing a Mule application to periodically poll an SFTP location for new files containing sales order records and then process those sales orders. Each sales order must be processed exactly once. To support this requirement, the Mule application must identify and filter duplicate sales orders on the basis of a unique ID contained in each sales order record and then only send the new sales orders to the downstream system. What is the most idiomatic (used for its intended purpose) Anypoint connector, validator, or scope that can be configured in the Mule application to filter duplicate sales orders on the basis of the unique ID field contained in each sales order record? A. Configure a Cache scope to filter and store each record from the received file by the order ID B. Configure an Idempotent Message Validator component to filter each record by the order ID C. Configure a Database connector to filter and store each record by the order ID D. Configure a watermark in an On New or Updated File event source to filter unique records by the order ID	B	https://docs.mulesoft.com/mule-runtime/4.4/idempotent-message-validator
74	The ACME Tractors organization has deployed both Mule and non-Mule API implementations to integrate its customer and order management systems. All the APIs are available to REST clients on the public internet. The organization wants to monitor these APIs by running health checks; for example, to determine if an API can properly accept and process requests. The organization does not have subscriptions to any external monitoring tools and also does not want to extend its IT footprint. What Anypoint Platform feature provides the most idiomatic (used for its intended purpose) way to monitor the availability of both the Mule and the non-Mule API implementations? A. Runtime Manager B. API Manager C. API Functional Monitoring D. Anypoint Visualizer	C	
75	A REST API is being designed. It will be implemented as a Mule application and managed on Anypoint Platform. What standard interface definition language can be used to define REST APIs? A. Web Services Description Language (WSDL) B. AsyncAPI Specification (AsyncAPI) C. OpenAPI Specification (OAS) D. YAML Ain't Markup Language (YAML)	C	https://docs.mulesoft.com/release-notes/platform/oas3

76	An external web UI application currently accepts occasional HTTP requests from client web browsers to change (insert, update, or delete) inventory pricing information in an inventory system's database. Each inventory pricing change must be transformed and then synchronized with multiple customer experience systems in near real-time (in under 10 seconds). New customer experience systems are expected to be added in the future. The database is used heavily and limits the number of SELECT queries that can be made to the database to 10 requests per hour per user. What is the most scalable, idiomatic (used for its intended purpose), decoupled, reusable, and maintainable integration mechanism available to synchronize each inventory pricing change with the various customer experience systems in near real-time? A. Write a Mule application with a Database On Table Row event source configured for the inventory pricing database, with the watermark attribute set to an appropriate database column In the same flow, use a Scatter-Gather to call each customer experience system's REST API with transformed inventory-pricing records B. Replace the external web UI application with a Mule application to accept HTTP requests from client web browsers In the same Mule application, use a Batch Job scope to test if the database request will succeed, aggregate pricing changes within a short time window, and then update both the inventory pricing database and each customer experience system using a Parallel For Each scope C. Add a trigger to the inventory-pricing database table so that for each change to the inventory pricing database, a stored procedure is called that makes a REST call to a Mule application Write the Mule application to publish each Mule event as a message to an Anypoint MQ exchange Write other Mule applications to subscribe to the Anypoint MQ exchange, transform each received message, and then update the Mule application's corresponding customer experience system(s) D. Write a Mule application with a Database On Table Row event source configured for the inventory pricing database, with the ID attribute set to an appropriate database column In the same flow, use a Batch Job scope to publish transformed inventory-pricing records to an Anypoint MQ queue Write other Mule applications to subscribe to the Anypoint MQ queue, transform each received message, and then update the Mule application's corresponding customer experience system(s)	A	
77	A Mule application uses APIkit for SOAP to implement a SOAP web service. The Mule application has been deployed to a CloudHub worker in a testing environment. The integration testing team wants to use a SOAP client to perform integration testing. To carry out the integration tests, the integration team must obtain the interface definition for the SOAP web service. What is the most idiomatic (used for its intended purpose) way for the integration testing team to obtain the interface definition for the deployed SOAP web service in order to perform integration testing with the SOAP client? A. Retrieve the XML file(s) from Runtime Manager B. Retrieve the RAML file(s) from the deployed Mule application C. Retrieve the OpenAPI Specification file(s) from API Manager D. Retrieve the WSDL file(s) from the deployed Mule application	D	https://www.infoviewsystems.com/creating-a-soap-api-via-mulesoft/
78	An organization is designing a Mule application to support an all-or-nothing transaction between several database operations and some other connectors so that they all automatically roll back if there is a problem with any of the connectors. Besides the Database connector, what other Anypoint connector can be used in the Mule application to participate in the all-or-nothing transaction? A. JMS(or VM) B. SFTP C. Anypoint MQ D. Object Store	A	
79	ACME Tractors has defined a common object model in Java to mediate the communication between different Mule applications in a consistent way. A Mule application is being built to use this common object model to process responses from a SOAP API and a REST API and then write the processed results to an order management system. The developers want Anypoint Studio to utilize these common objects to assist in creating mappings for various transformation steps in the Mule application. What is the most idiomatic (used for its intended purpose) and performant way to utilize these common objects to map between the inbound and outbound systems in the Mule application? A. Use the WSS module B. Use the Java module C. Use JAXB (XML) and Jackson (JSON) data bindings D. Use the Transform Message component	D	
80	The ACME Best Insurance company has an Anypoint Runtime Fabric on VMs/Bare Metal (RTF-VM) appliance installed on its own customer-hosted AWS infrastructure. Mule applications are deployed to this RTF-VM appliance. As part of the company standards, the Mule application logs must be forwarded to an external log management tool (LMT). Given the company's current setup and requirements, what is the most idiomatic (used for its intended purpose) way to send Mule application logs to the external LMT? A. In RTF-VM, install and configure the external LTM's log-forwarding agent B. In RTF-VM, configure the out-of-the-box external log forwarder C. In RTF-VM, edit the pod configuration to automatically install and configure an Anypoint Monitoring agent D. In each Mule application, configure custom Log4j settings	A	https://docs.mulesoft.com/runtime-fabric/2.1/#external-log-forwarding

81	ACME Tractors has an HTTPS-enabled Mule application named Orders API that receives requests from another Mule application named Process Orders. The communication between these two Mule applications must be secured by TLS mutual authentication (two-way TLS). At a minimum, what must be stored in each truststore and keystore of these two Mule applications to properly support two-way TLS between the two Mule applications while properly protecting each Mule application's keys? A. Orders API truststore: The Process Orders public key Orders API keystore: The Orders API private key Process Orders truststore: The Orders API public key Process Orders keystore: The Process Orders private key B. Orders API truststore: The Orders API private key and public key Process Orders keystore: The Process Orders private key and public key C. Orders API truststore: The Process Orders public key Orders API keystore: The Orders API private key and public key Process Orders truststore: The Orders API public key Process Orders keystore: The Process Orders private key and public key D. Orders API truststore: The Orders API public key Process Orders keystore: The Process Orders private key and public key	C	
82	In Anypoint Platform, a company wants to configure multiple identity providers (IdPs) for multiple lines of business (LOBs). Multiple business groups, teams, and environments have been defined for these LOBS. What Anypoint Platform feature can use multiple IdPs across the company's business groups, teams, and environments? A. Client Management B. Roles and permissions C. MuleSoft-hosted (CloudHub) dedicated load balancers D. Virtual private clouds	A	https://docs.mulesoft.com/access-management/environments#add-an-idp-to-an-existing-environment
83	A Mule application is being designed to perform product orchestration. The Mule application must join together the responses from an Inventory API and a Product Sales History API with the least latency. To minimize the overall latency, what is the most idiomatic (used for its intended purpose) design to call each API request in the Mule application as quickly as possible? A. Call each API request in a separate Async scope B. Call each API request in a separate route of a Scatter-Gather C. Call each API request in a separate lookup call from a DataWeave reduce function D. Call each API request in a separate route of a Parallel For Each scope	B	
84	An ACME Farms project team is planning to build a new API that is required to work with data from different domains across the organization. The organization has a policy that all project teams should leverage existing investments by reusing existing APIs and related resources and documentation that other project teams have already developed and deployed. To support reuse, where on Anypoint Platform should the project team go to discover and read existing APIs, discover related resources and documentation, and interact with mocked versions of those APIs? A. Design Center B. API Manager C. Anypoint Exchange D. Runtime Manager	C	https://docs.mulesoft.com/design-center/design-mocking-service
85	ACME Tractors is designing an integration Mule application to process orders by submitting them to a back-end system for offline processing. Each order will be received by the Mule application through an HTTPS POST and must be acknowledged immediately. Once acknowledged, the order will be submitted to a back-end system. Orders that cannot be successfully submitted due to rejections from the back-end system will need to be processed manually (outside the back-end system). The Mule application will be deployed to a customer-hosted runtime and is able to use an existing ActiveMQ broker if needed. The ActiveMQ broker is located inside the organization's firewall. The back-end system has a track record of unreliability due to both minor network connectivity issues and longer outages. What idiomatic (used for their intended purposes) combination of Mule application components and ActiveMQ queues are required to ensure automatic submission of orders to the back-end system while supporting but minimizing manual order processing? A. One or more On Error scopes to assist calling the back-end system One or more ActiveMQ long-retry queues A persistent dead-letter object store configured in the CloudHub Object Store service B. A Batch Job scope to call the back-end system An Until Successful scope containing Object Store components for long retries A dead-letter object store configured in the Mule application C. An Until Successful scope to call the back-end system One or more ActiveMQ long-retry queues One or more ActiveMQ dead-letter queues for manual processing D. One or more On Error scopes to assist calling the back-end system An Until Successful scope containing VM components for long retries A persistent dead-letter VM queue configured in CloudHub	C	